

Ethan Arnold NE 533 Exam 2

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1. $r = 0.55 \text{ cm}$, $LHR = 300 \frac{\text{W}}{\text{cm}}$, $k_c = 0.025 \frac{\text{W}}{\text{cm K}}$, $E = 200 \text{ GPa}$, $\nu = 0.35$
 $\alpha = 10 \times 10^{-6} \frac{1}{\text{K}}$

a) $\sigma^* = \frac{\alpha E (T_o - T_s)}{4(1-\nu)}$, $\Delta T = T_o - T_s = \frac{LHR}{4\pi k_c}$

$\Delta T = \frac{300 \frac{\text{W}}{\text{cm}}}{4\pi (0.025 \frac{\text{W}}{\text{cm K}})} \rightarrow \Delta T = 954.93 \text{ K}$

$\sigma^* = \frac{(10 \times 10^{-6} \frac{1}{\text{K}})(200 \text{ GPa})(954.93 \text{ K})}{4(1-0.35)} \rightarrow \sigma^* = 0.735 \text{ GPa or } 735 \text{ MPa}$

$\nu = \frac{r}{R_f}$, $\sigma_r(0) = -735(1-0^2) = -735 \text{ MPa}$, $\sigma_r(1) = -735(1-1^2) = 0$
 $\sigma_\theta(0) = -735(1-3(0)^2) = -735 \text{ MPa}$, $\sigma_\theta(1) = -735(1-3(1)^2) = 1470 \text{ MPa}$
 $\sigma_z(0) = -2(735)(1-2(0)^2) = 1470 \text{ MPa}$, $\sigma_z(1) = -2(735)(1-2(1)^2) = 1470 \text{ MPa}$

$\sigma_\theta(\text{max})$ at $\nu=1$, $r=0.55 \text{ cm}$ of 1470 MPa , $\sigma_z = 1470 \text{ MPa}$

b) $\sigma_{fr} = 175 \text{ MPa} = \sigma_\theta(\nu) + 175 \text{ MPa} = -735 \text{ MPa}(1-3(\nu)^2)$ - some with wrong
 $1-3\nu^2 = 0.238 \rightarrow 3\nu^2 = 0.762 \rightarrow \nu^2 = 0.254 \rightarrow \nu = 0.504$

$0.504 = \frac{r}{0.55 \text{ cm}} \rightarrow r = 0.277 \text{ cm}$ or 49.6% of the fuel has cracks

- right process

2. $p = 15 \text{ MPa}$, $\bar{r} = 0.55 \text{ cm}$, $t = 0.05 \text{ cm}$, $R_i = 0.5 \text{ cm}$, $R_o = 0.6 \text{ cm}$
 $\frac{R_o}{R_i} = 1.2$, $r = 0.5 \text{ cm}$

a) $\sigma_\theta = \frac{PR}{t} = \frac{15 \text{ MPa}(0.55 \text{ cm})}{0.05 \text{ cm}} \rightarrow \sigma_\theta = 165 \text{ MPa}$
 $\sigma_z = \frac{PR}{2t} = \frac{15 \text{ MPa}(0.55 \text{ cm})}{2(0.05 \text{ cm})} + \sigma_z = \frac{15 \text{ MPa}(0.55 \text{ cm})}{2(0.05 \text{ cm})} = 82.5 \text{ MPa} = \sigma_z$
 $\sigma_r = -\frac{P}{2} = \frac{15 \text{ MPa}}{2} \rightarrow \sigma_r = -7.5 \text{ MPa}$

b) $\sigma_r = -15 \text{ MPa} \frac{(0.6/0.5)^2 - 1}{(1.2)^2 - 1} \rightarrow \sigma_r = -15 \text{ MPa}$
 $\sigma_\theta = 15 \text{ MPa} \frac{(1.2)^2 + 1}{(1.2)^2 - 1} \rightarrow \sigma_\theta = 83.18 \text{ MPa}$

$\sigma_z = 15 \text{ MPa} \frac{1}{(1.2)^2 - 1} \rightarrow \sigma_z = 34.1 \text{ MPa}$

$R_i = 0.525$
 $R_o = 0.575$

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3. $R_f = 0.48 \text{ cm}$, $t_{\text{gap}} = 0.008 \text{ cm}$, $T_{10} = 550 \text{ K}$, $t_{\text{chd}} = 0.08 \text{ cm}$, $k_{\text{fuel}} = 0.03 \frac{\text{W}}{\text{cm K}}$
 $k_{\text{gap}} = 0.003 \frac{\text{W}}{\text{cm K}}$, $k_{\text{chd}} = 0.15 \frac{\text{W}}{\text{cm K}}$, $LHR = \frac{250}{\text{W/cm}}$, $\alpha_c = 10 \times 10^{-6} \frac{1}{\text{K}}$
 $\alpha_f = 14 \times 10^{-6} \frac{1}{\text{K}}$, $T_{\text{ref}} = 300 \text{ K}$

$$T_{1c} = T_{10} + \frac{LHR}{2\pi R_f h_{\text{chd}}}, h_{\text{chd}} = \frac{k_{\text{chd}}}{t_{\text{chd}}} = \frac{0.15 \frac{\text{W}}{\text{cm K}}}{0.08 \text{ cm}} = 1.875 \frac{\text{W}}{\text{cm}^2 \text{ K}}$$

$$T_{1c} = 550 \text{ K} + \frac{250 \frac{\text{W/cm}}{2\pi (0.48 \text{ cm}) (1.875 \frac{\text{W}}{\text{cm}^2 \text{ K}})} \rightarrow T_{1c} = 594.21 \text{ K}$$

$$T_s = T_{1c} + \frac{LHR}{2\pi R_f h_{\text{gap}}}, h_{\text{gap}} = \frac{k_{\text{gap}}}{t_{\text{gap}}} = \frac{0.003 \frac{\text{W}}{\text{cm K}}}{0.008 \text{ cm}} \rightarrow h_{\text{gap}} = 0.375 \frac{\text{W}}{\text{cm}^2 \text{ K}}$$

$$T_s = 594.21 + \frac{250 \frac{\text{W/cm}}{2\pi (0.48 \text{ cm}) (0.375 \frac{\text{W}}{\text{cm}^2 \text{ K}})} \rightarrow T_s = 815.26 \text{ K}$$

$$T_0 = T_s + \frac{LHR}{4\pi k_f} \rightarrow T_0 = 815.26 \text{ K} + \frac{250 \frac{\text{W/cm}}{4\pi (0.03 \frac{\text{W}}{\text{cm K}})} \rightarrow T_0 = 1478.41 \text{ K}$$

$$\bar{T}_c = \frac{1478.41 + 815.26}{2} = 1146.83 \text{ K}, \bar{T}_c = \frac{(594.21 \text{ K} + 550 \text{ K})}{2} = 572.11 \text{ K}$$

$$\bar{R}_c = 0.48 + 0.008 + \frac{0.08}{2} = 0.528$$

$$\Delta \delta_{\text{gap}} = (0.528 \text{ cm}) (10 \times 10^{-6} \frac{1}{\text{K}}) (572.11 \text{ K} - 300 \text{ K}) - (0.48 \text{ cm}) (14 \times 10^{-6} \frac{1}{\text{K}}) (1146.83 \text{ K} - 300 \text{ K})$$

$$\Delta \delta_{\text{gap}} = -0.00425 \text{ cm} \quad \delta_{\text{gap,ref}} = 0.003746$$

4. $\Delta T = 50 \text{ K}$, $\alpha_c = 15 \times 10^{-6}$, $E = 100 \text{ GPa}$, $\nu = 0.34$, $t_c = 0.06 \text{ cm}$, $R_c = 0.55 \text{ cm}$, $R_0 = 0.61 \text{ cm}$

$$\sigma_r = \frac{1}{2} (50 \text{ K}) (15 \times 10^{-6} \frac{1}{\text{K}}) (100 \text{ GPa}) \left(\frac{0.55}{0.55 - 1} \right) \left(1 - \frac{0.55}{0.06} \left(\frac{0.55}{0.55} - 1 \right) \right)$$

$$\sigma_r = 0 \quad (1 - 0.34)$$

$$\sigma_\theta = \frac{1}{2} (50 \text{ K}) (15 \times 10^{-6} \frac{1}{\text{K}}) (100 \text{ GPa}) \left(1 - 2 \frac{0.55}{0.06} \left(\frac{0.55}{0.55} - 1 \right) \right) (1 - 0.34)$$

$$\sigma_\theta = 0.0568 \text{ GPa} \quad \text{or} \quad 56.8 \text{ MPa} = \sigma_\theta$$

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5. $\dot{F} = 6 \times 10^{13} \text{ dis/cm}^2\text{-s}$, $\Delta f_0 = 0.015$, $B_0 = 5 \frac{\text{MWd}}{\text{kgU}}$, $p(\text{UO}_2) = 10.97 \text{ g/cc}$
 $t = 10 \text{ days}$ Assume $\phi = 1$

Assuming 270 $\frac{\text{g}}{\text{mol}}$ for UO_2 since enrichments not given

$$10.97 \frac{\text{g}}{\text{cc}} \times \frac{1 \text{ mol UO}_2}{270 \text{ g}} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol}} \times \frac{1 \text{ U}}{1 \text{ UO}_2} = 2.447 \times 10^{22} \frac{\text{U}}{\text{cc}}$$

$$B_0 = \frac{5 \text{ MWd/kgU}}{950} \rightarrow B_0 = 0.0053 \text{ FI/Ma}$$

$$P = \frac{(6 \times 10^{13} \text{ dis/cm}^2\text{-s}) (10 \text{ days} \times 24 \frac{\text{hr}}{\text{day}} \times \frac{3600 \text{ s}}{\text{hr}})}{2.447 \times 10^{22} \frac{\text{U}}{\text{cc}}} = 0.0021 \text{ FI/Ma}$$

$$\epsilon_0 = 0.015 \left(e^{\left(\frac{0.0021 \ln(0.01)}{1 - 0.0053} \right)} - 1 \right) \rightarrow \boxed{\epsilon_D = 0.0089}$$

6. Elasticity is the ability of a material to resist deformation in a way that allows it to regain its shape when the force is removed (temporary deformation). Plasticity is the permanent deformation of a material from a force or load resulting from the movement of dislocations (permanent).

7. Strain hardening is the tendency of a material's yield strength to increase as it plastically deforms, when a material plastically deforms and a load is released and reapplied, the material's yield strength will increase. This is caused by the pile up of dislocations along grain boundaries which resists plastic strain.

8. 1) give the temperature profile / swelling, 2) stresses in the clad
 3) model the gap (FGR and other properties)

9. Grain boundaries form as particles are fused together during the sintering process. These particles are oriented differently and create grain boundaries due to differences in inclination and misorientation.

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10. Microstructure-based (mechanistic) fuel performance modelling uses changes in the microstructure to determine the effects on the material properties rather than relying on empirical burnup correlations. This can be a more accurate model since it accounts for grain size, precipitates, defects, etc. It can also be less expensive since empirical correlations often require significant data collection.

11. The microstructure is meant by that which is microscopically visible (25x) and includes the grains, grain boundaries, grain size, texture, orientations, impurities, voids, precipitates and other material effects on the microscopic scale of crystalline materials. Quenching is used to heat up a material and quickly cool it to lock in the high temperature microstructure. Rolling may also be used to force dislocation movement and pileup for a harder material.

12. Diffusion occurs more quickly on the grain boundaries.

Inter > Intra - why?

13. The driving force for grain growth is the reduction in grain boundary energy called the curvature driving force. Solutes at the grain boundary can impede grain growth known as solute drag.

14. Fuel densifies because it is not 100% dense to begin with. There are smaller (and sometimes larger) pores that exist within the structure. Densification is effectively a continuation of sintering that closes these smaller pores. Irradiation can speed this up. Larger pores are often more stable unless there is a hydrostatic stress. Diffusion of voids contributes to this process and densification overall is due to lowering surface area energy.